

DATE : 18.11.2025

DT/NT : DT

LESSON : STATISTICS I

**SUBJECT: PARAMETER AND STATS,
GRAPHICAL REPRESENTATION**

BATCH : 322

**DATA
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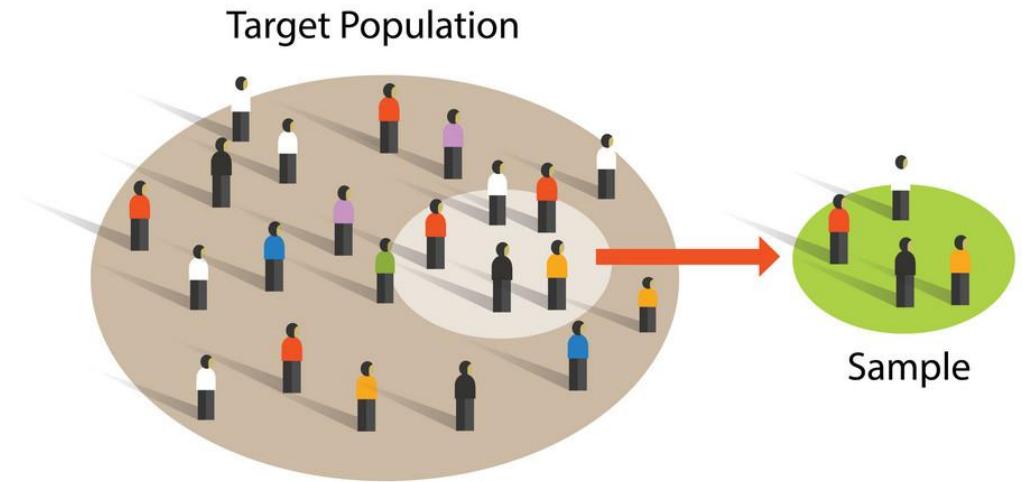
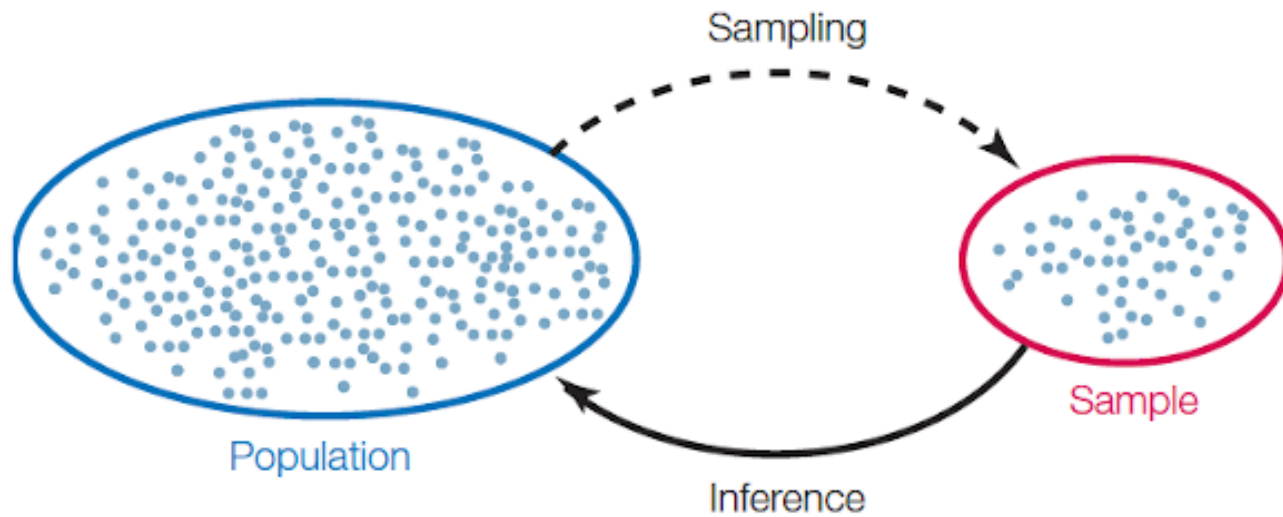


Can you write a sentence that you remember from the previous lesson?



**Have you practiced
with the **pre-class**
materials that will
prepare you for
today's lesson?**

7. Parameters and Statistics

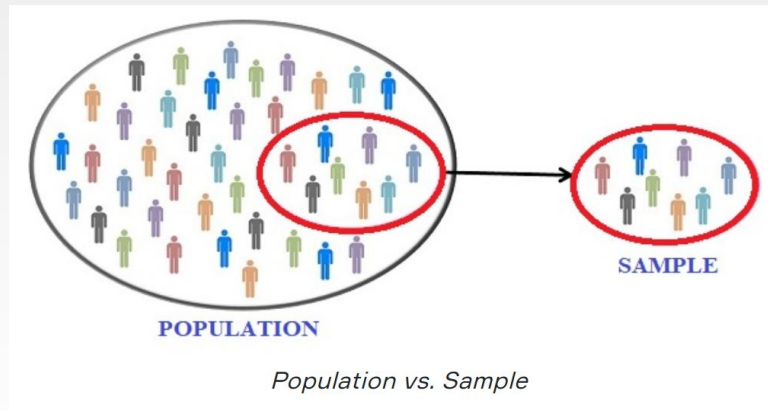


7. Populations & Samples

➤ Population

➤ Sample

- A statistical study is based entirely on the examination of the data set or solution space.



!!! We observe samples but we are interested in populations.

7. Parameters & Statistics

**Population
Attributes**



Parameters

Sample Attributes

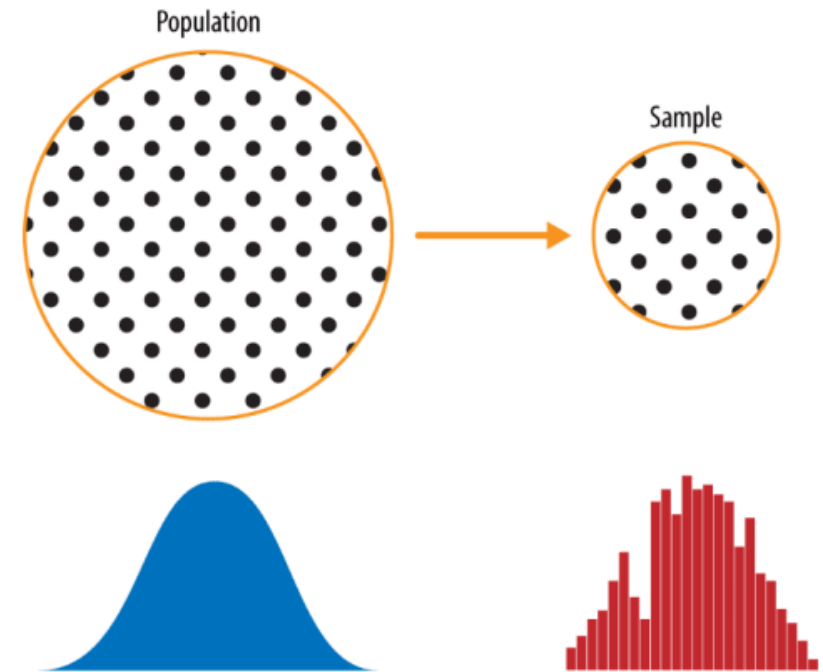
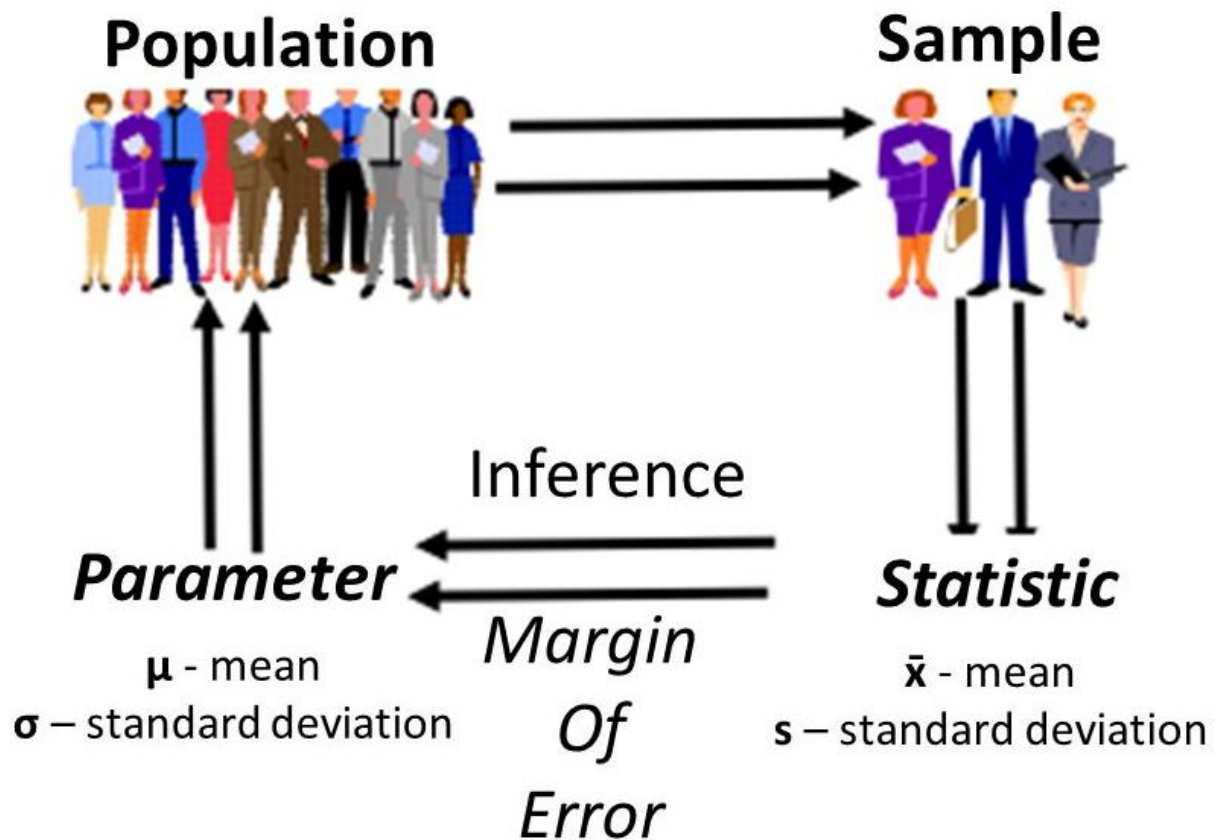


Statistics

A parameter is a numerical summary of the population, and a statistic is a numerical summary of a sample.

7. Parameters and Statistics

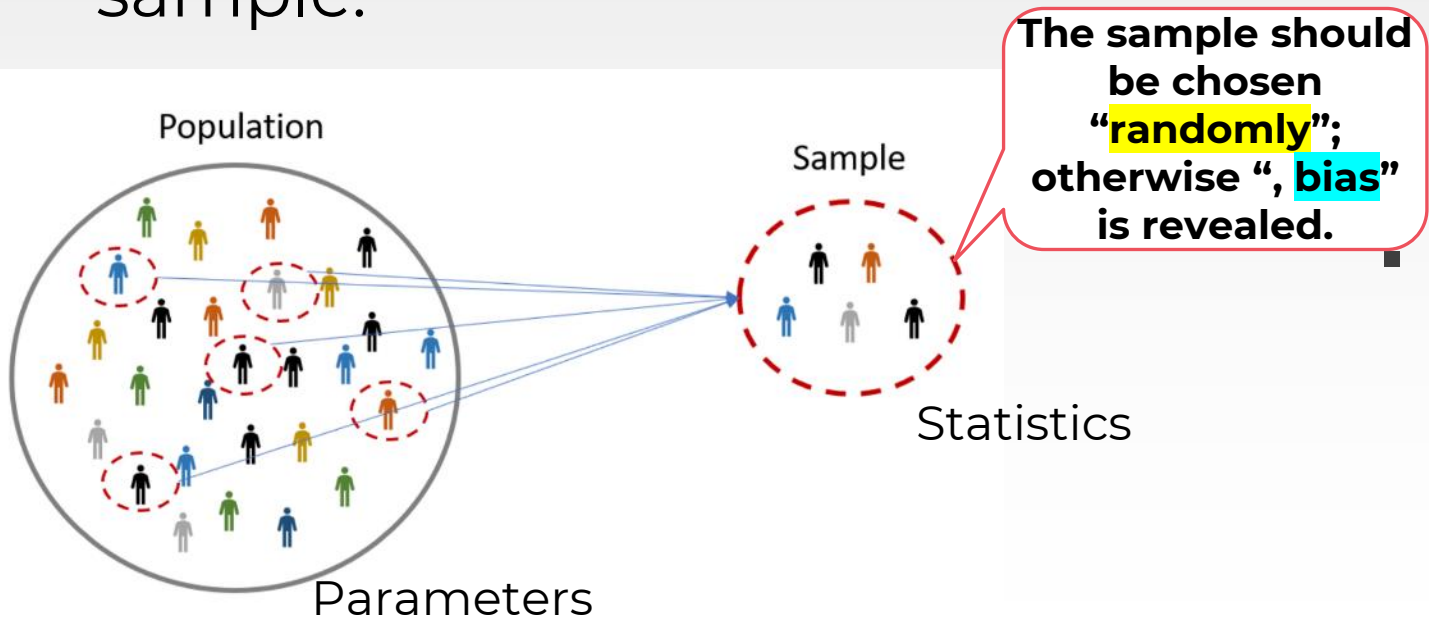
The Basic Paradigm



- Populations have Parameters (like μ , σ^2 , θ , p)
- Samples have Statistics, functions of observed data, like $\bar{x}$, $\tilde{x}$, s^2 , $\hat{\theta}$, $\hat{p}$

7. Parameters and Statistics

- A parameter is a numerical description of the characteristics of the population.
- A statistic is a numerical description of the characteristics of the sample.



Sample (statistic):

- Let us decide whether the given value describes a population parameter or a sample statistic.
- a.) A survey of a sample of 450 college students reports that the average weekly income for students is \$325.
 - Since the average of \$325 is based on a sample, this is a sample statistic.
- b.) The average weekly income for all students is 405 dollars.
 - Since the average of \$405 is based on a population, this is a population parameter.

Top 60 Statistics Interview Questions 2024



Question 2: What is Random Sampling? Give some examples of some random sampling techniques.

Answer: Random sampling is a **sampling method** in which **each sample has an equal probability of being chosen as a sample**. It is also known as probability sampling.

Top 60 Statistics Interview Questions 2024



Question 3: What is the difference between population and sample in inferential statistics?

Answer: A **population** in inferential statistics **refers to the entire group** we take samples from and are used to draw conclusions. A **sample**, on the other hand, **is a specific group we take data from population**. This data is used to calculate the statistics. **Sample size is always less than that of the population.**

9. Level of Measurement

- In the **1940s, Stanley Smith Stevens** introduced four levels of measurement - scales: nominal, ordinal, interval and ratio.
- These are still widely used today as **a way of describing the characteristics of a variable**. Knowing the level-scale of measurement of a variable **is an important consideration in choosing the right statistical analysis**.

LEVEL OF MEASUREMENT	
TYPE	MEASURE PROPERTY
NOMINAL	Classification, membership $=, \neq$
ORDINAL	Comparison Level $>, <$
INTERVAL	Difference, closeness $+, -$
RATIO	Magnitude, amount $*, /$

9. Level of Measurement

NOMINAL DATA

Nominal data divides variables into mutually exclusive, labeled categories.

Examples

Eye color



Smartphone



Transport



How is nominal data analyzed?

Descriptive statistics: Frequency distribution and mode

Non-parametric statistical tests

ORDINAL DATA

Ordinal data classifies variables into categories which have a natural order or rank.

Examples

School grades



Education level



Seniority level



How is ordinal data analyzed?

Descriptive statistics: Frequency distribution, mode, median, and range

Non-parametric statistical tests

INTERVAL DATA

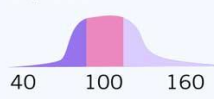
Interval data is measured along a numerical scale that has equal intervals between adjacent values.

Examples

Temperature



IQ score



Income ranges



How is interval data analyzed?

Descriptive statistics: Frequency distribution; mode, median, and mean; range, standard deviation, and variance

Parametric statistical tests (e.g. t-test, linear regression)

RATIO DATA

Ratio data is measured along a numerical scale that has equal distances between adjacent values, and a true zero.

Examples

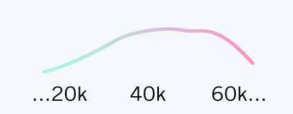
Weight in KG



Number of staff



Income in USD

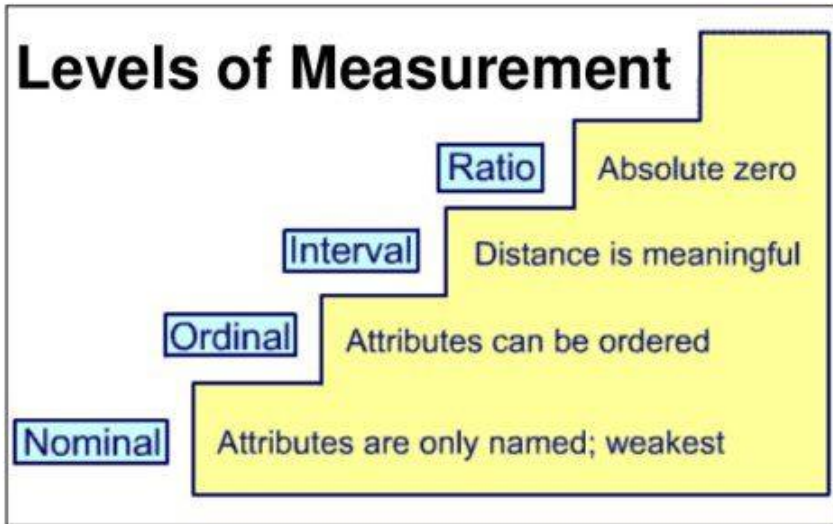


How is ratio data analyzed?

Descriptive statistics: Frequency distribution; mode, median, and mean; range, standard deviation, variance, and coefficient of variation

Parametric statistical tests (e.g. ANOVA, linear regression)

9. Level of Measurement



THE FOUR LEVELS OF MEASUREMENT:

	Nominal	Ordinal	Interval	Ratio
Categorizes and labels variables	✓	✓	✓	✓
Ranks categories in order		✓	✓	✓
Has known, equal intervals			✓	✓
Has a true or meaningful zero				✓

Differences between measurements, true zero exists

Ratio Data

Quantitative Data

Differences between measurements but no true zero

Interval Data

Ordered Categories (rankings, order, or scaling)

Ordinal Data

Qualitative Data

Categories (no ordering or direction)

Nominal Data

Levels of Measurement

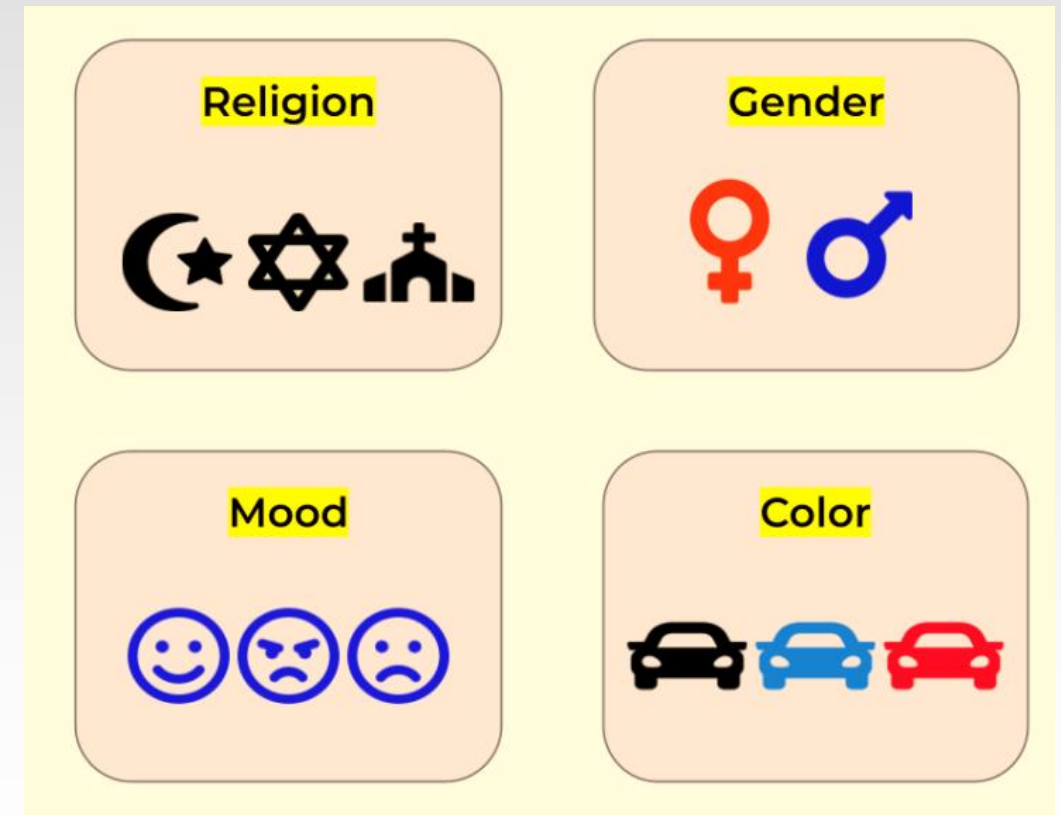
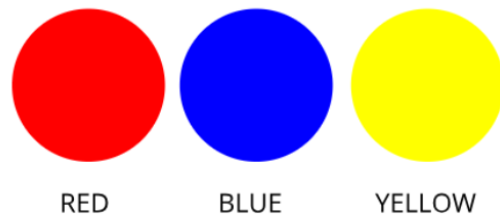
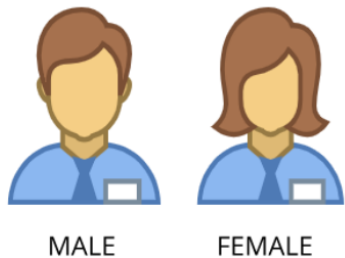
Nominal	Ordinal	Interval	Ratio
"Eye color"	"Level of satisfaction"	"Temperature"	"Height"
Named	Named	Named	Named
	Natural order	Natural order	Natural order
		Equal interval between variables	Equal interval between variables
			Has a "true zero" value, thus ratio between values can be calculated

9. Level of Measurement

► Nominal Level of Measurement

Also called classificatory level of measurement

- No ranking
- There is categorization
- There is belonging



9. Level of Measurement

NOMINAL DATA

Nominal data divides variables into mutually exclusive, labeled categories.

Examples

Eye color



Smartphone



Transport



How is nominal data analyzed?

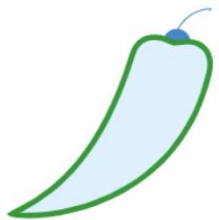
Descriptive statistics:
Frequency distribution
and mode

**Non-parametric
statistical tests**

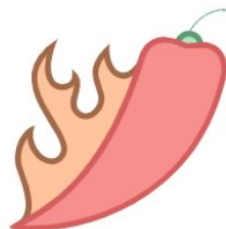
9. Level of Measurement

➤ Ordinal Level of Measurement

- There is a ranking
- No real value
- Comparison available



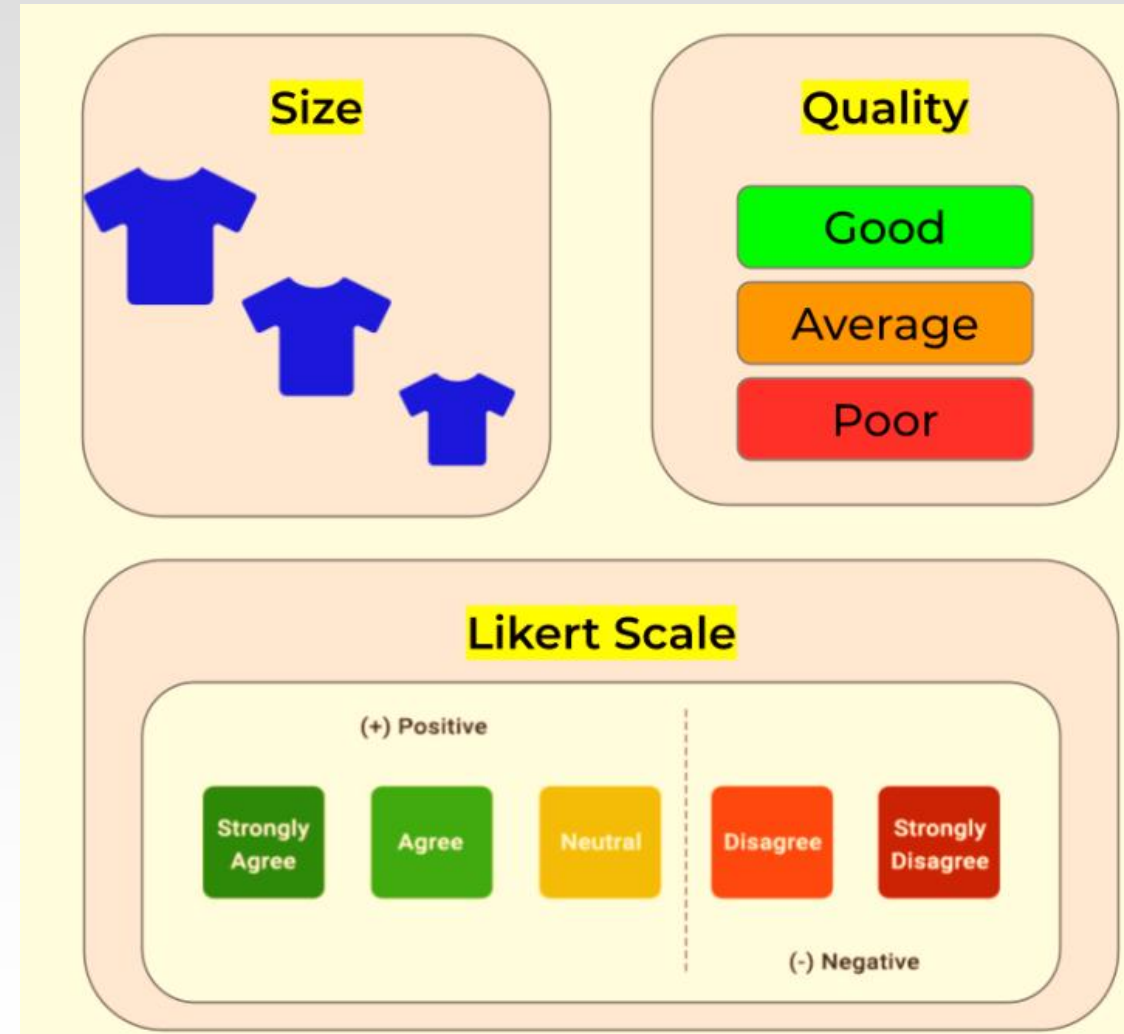
HOT



HOTTER



HOTTEST



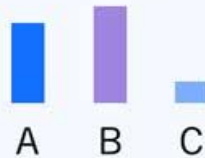
9. Level of Measurement

ORDINAL DATA

Ordinal data classifies variables into categories which have a natural order or rank.

Examples

School grades



Education level



Seniority level



How is ordinal data analyzed?

Descriptive statistics:
Frequency distribution,
mode, median, and range

**Non-parametric
statistical tests**

Types of Measurement Scales

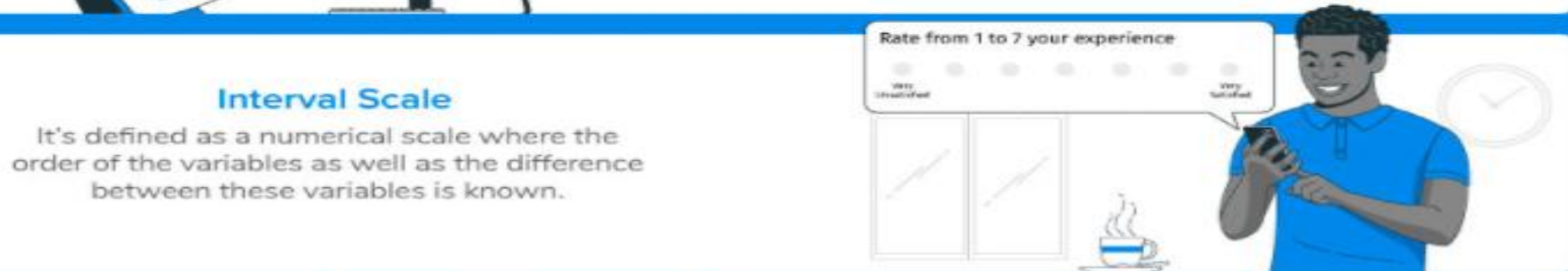
Nominal scale

It's used to label variables in different classifications and does not imply a quantitative value or order.



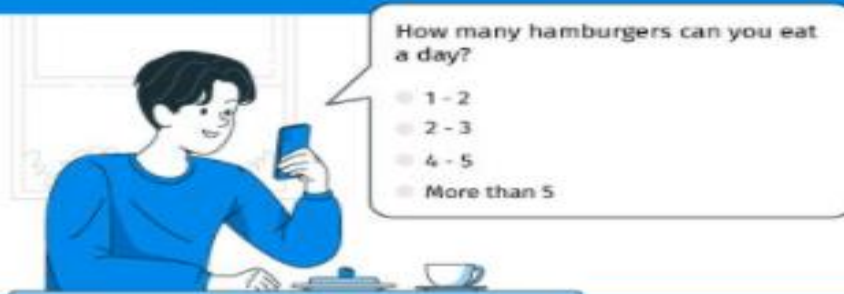
Ordinal Scale

It's used to represent non-mathematical ideas such as frequency, satisfaction, happiness, a degree of pain, etc.



Interval Scale

It's defined as a numerical scale where the order of the variables as well as the difference between these variables is known.



Ratio Scale

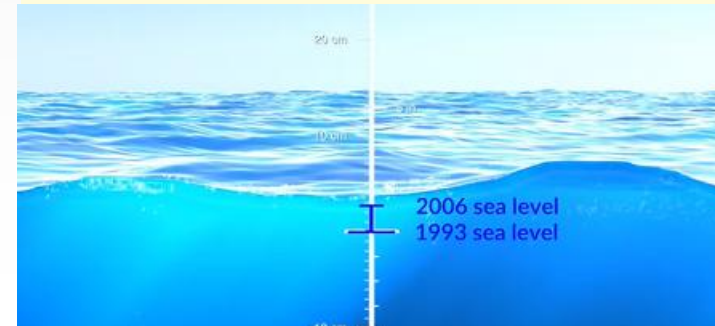
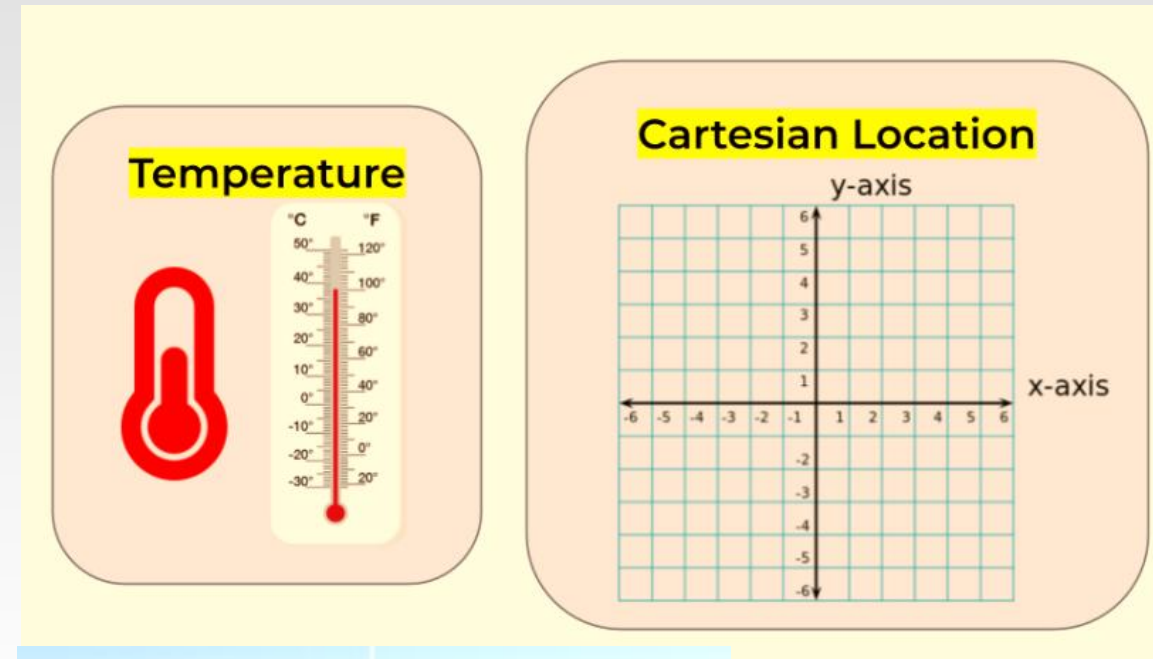
It's a variable measurement scale that not only produces the order of the variables, but also makes the difference between the known variables along with information about the value of the true zero.

9. Level of Measurement

► Interval Level of Measurement

Also called **equal interval measurement level**

- Measurable
- There is no true zero, there is **arbitrary zero**
- There are expressions of greatness, but they are not multiples of each other.
- E.g. $20^{\circ}\text{C} \neq 2 * 10^{\circ}\text{C}$.
- **Altitude above sea level** $2\text{m} \neq 2 * 1\text{m}$



9. Level of Measurement

INTERVAL DATA

Interval data is measured along a numerical scale that has equal intervals between adjacent values.

Examples

Temperature



IQ score



Income ranges



How is interval data analyzed?

Descriptive statistics: Frequency distribution; mode, median, and mean; range, standard deviation, and variance

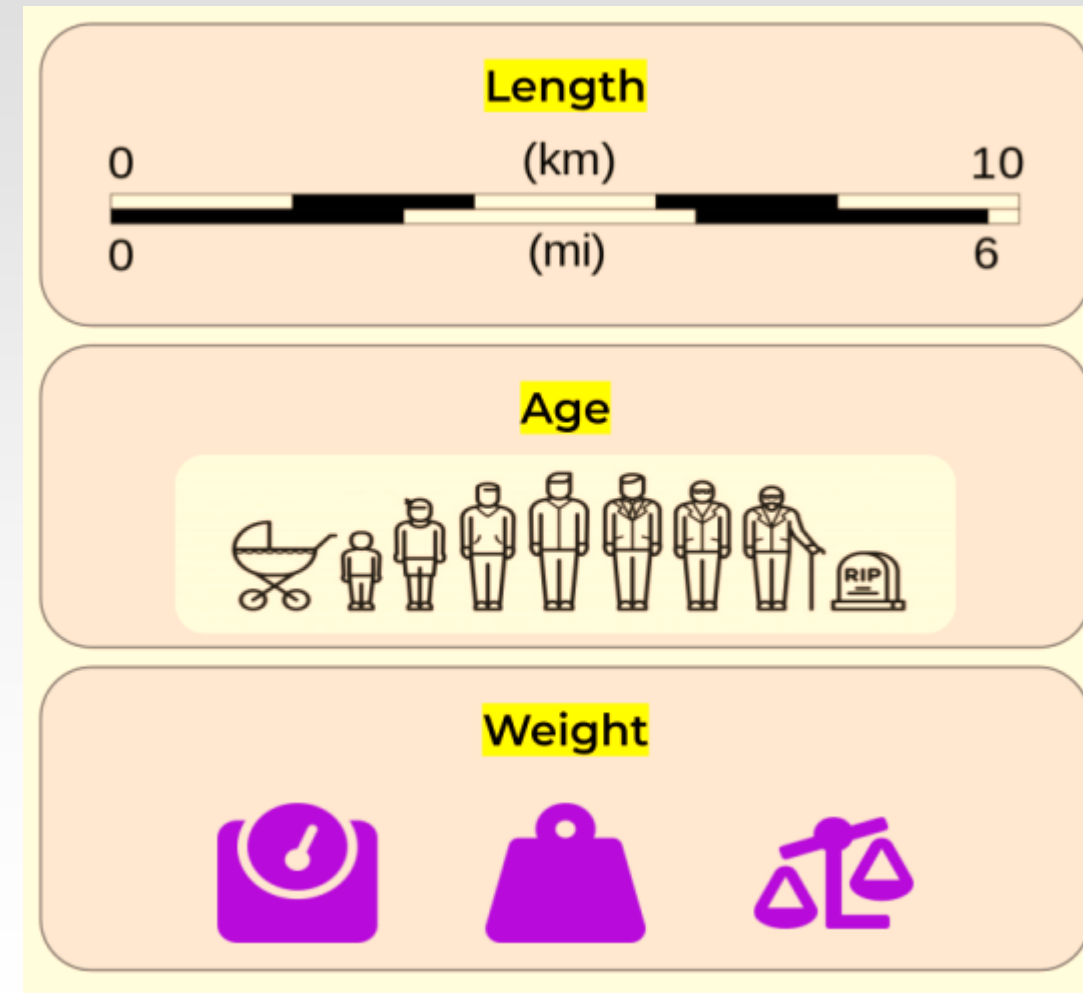
Parametric statistical tests (e.g. t-test, linear regression)

9. Level of Measurement

► Ratio Level of Measurement

Also called proportional measurement level

- Measurable
- Has True/Absolute zero
- There is an expression of greatness



9. Level of Measurement

RATIO DATA

Ratio data is measured along a numerical scale that has equal distances between adjacent values, and a true zero.

Examples

Weight in KG



Number of staff



Income in USD



How is ratio data analyzed?

Descriptive statistics: Frequency distribution; mode, median, and mean; range, standard deviation, variance, and coefficient of variation

Parametric statistical tests (e.g. ANOVA, linear regression)

9. Level of Measurement (Operational Level)

Categorical Data



Nominal

No rank or ordering

Colours, Days

Ordinal

Natural order or rank

Hot, Hotter, Hottest

Numerical Data



Interval

Measurable, but arbitrary zero

Altitude sea level
IQ level
Magnitude of earthquake

Ratio

Measurable and absolute zero

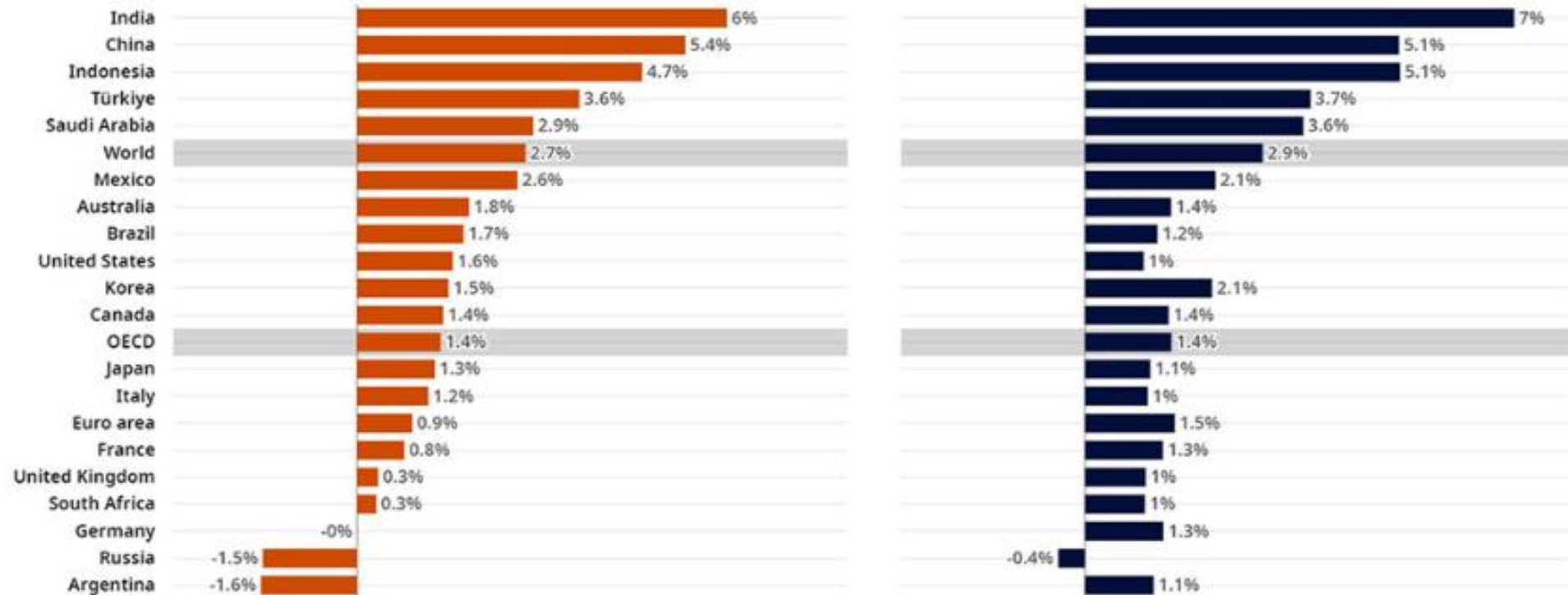
Income (\$)
Weight, Length

QUESTION

What kind of Level of Measurement is Economic Growth?

Real GDP growth projections for 2023 and 2024 %, year-on-year

Major economies OECD countries



Source: OECD Economic Outlook, June 2023.



Matching Variable vs. Level of Measurement

School grade

Nominal

Temperature

Ordinal

Weight

Interval

Eye Color

Ratio

Length

Task -1

EXERCISE 1. At what level are each of the following variables operationalized?

	Variable Name	Value Labels	Data Type
1	Age	1, 2, 3,	<i>Continuous</i>
2	Sex	male, female, other	
3	Class standing	freshman, sophomore, junior, senior	
4	Marital Status	married, single, divorced, widowed	
5	Median Household Income	annual household income in dollars	
6	How much you like the food on campus?	a lot, some, a little, other	
7	The number of toxic waste sites in your community	0, 1, 2, 3,	
8	The number of toxic waste sites in your community	0, 1–5, 6–10, 11+	
9	Your GPA	below average, average, above average	

Task -1

Variable Name	Value Labels	Data Type	Operational Level
Age	1, 2, 3,	Continuous	Ratio
Sex	male, female, other	Nominal	Nominal
Class standing	freshman, sophomore, junior, senior	Ordinal	Ordinal
Marital Status	married, single, divorced, widowed	Nominal	Nominal
Median Household Income	annual household income in dollars	Continuous	Ratio
How much you like the food on campus?	a lot, some, a little, other	Ordinal	Ordinal
The number of toxic waste sites in your community (exact number)	0, 1, 2, 3,	Discrete	Ratio
The number of toxic waste sites in your community (ranges)	0, 1–5, 6–10, 11+	Ordinal	Ordinal
Your GPA	below average, average, above average	Ordinal	Ordinal

It is time to Kahoot !!

Kahoot!

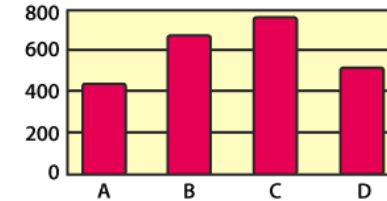
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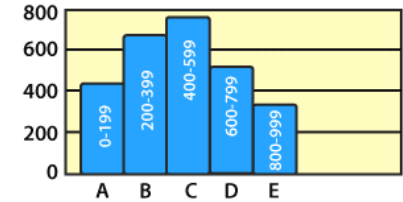
Data Visualization - Graphical Represent

- Graphical Representation of Data
 - Center
 - Spread
 - Shape
 - Unusual Features

Bar Graphs



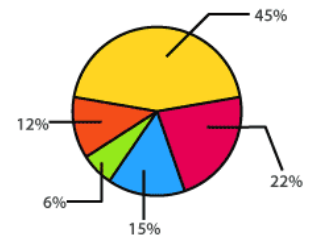
Histograms



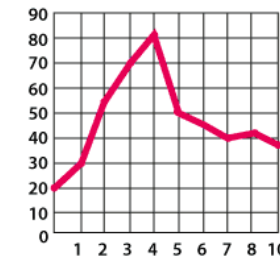
Frequency Table

Rulers of France		
Reign (Years)	Tally	Frequency
1-15		18
16-30		11
31-45		6
46-60		4
61-75		1

Circle Graph



Line Graphs

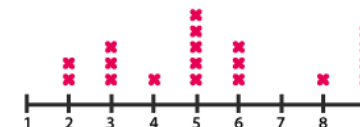


Stem and Leaf Plot

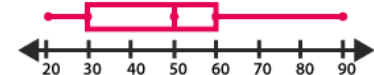
Stem	Leaf
0	1, 1, 2, 2, 3, 4, 4, 4, 4, 5, 8
1	0, 0, 0, 1, 1, 3, 7, 9
2	5, 5, 7, 7, 8, 8, 9, 9
3	0, 1, 1, 1, 2, 2, 2, 4, 5
4	0, 4, 8, 9
5	2, 6, 7, 7, 8
6	3, 6

Key : 6 | 3 = 63 Year

Line Plot



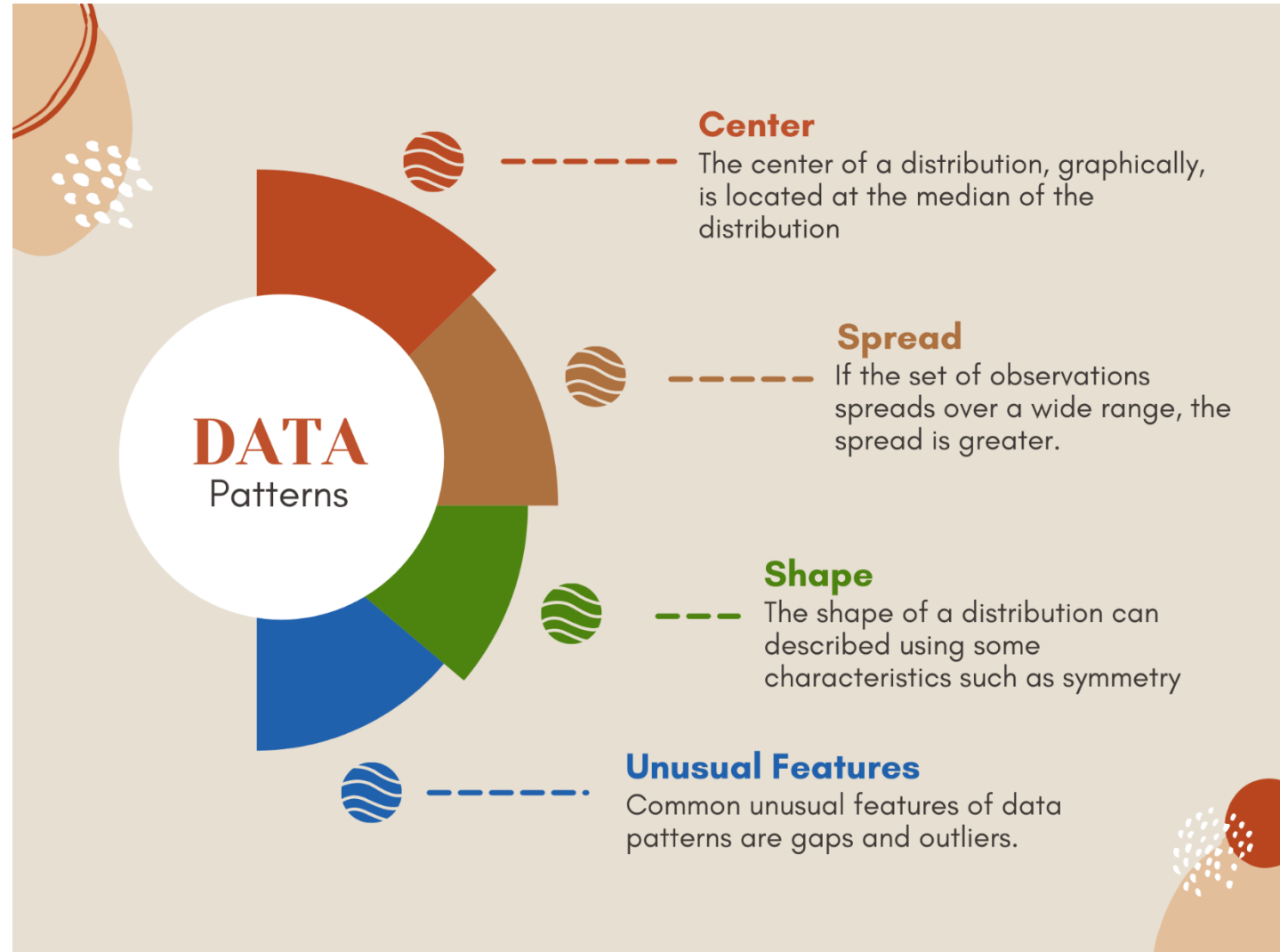
Box and Whisker Plot



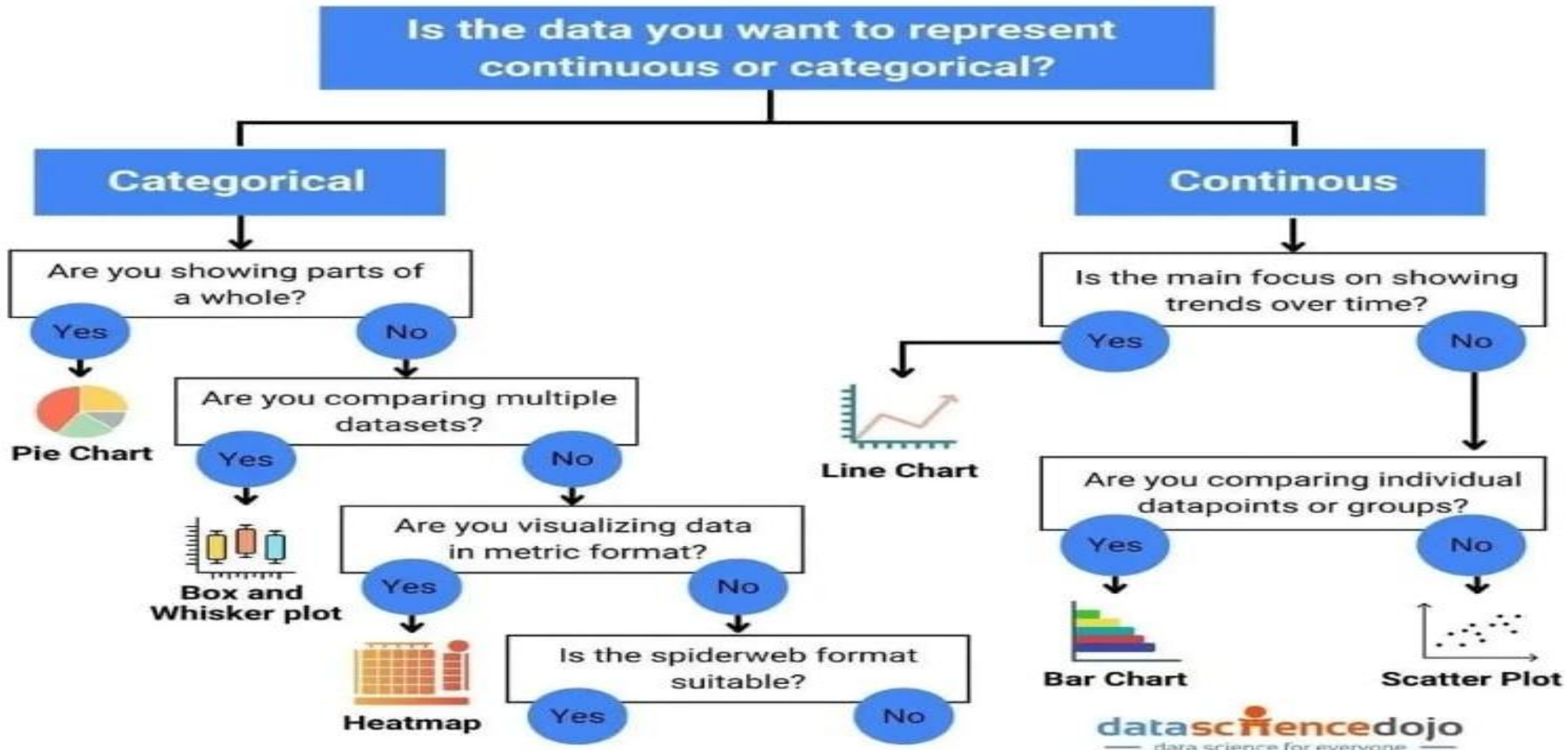
Data Patterns

- Data Patterns

- Center
- Spread
- Shape
 - Symmetric
 - Number of peaks
 - Skewness
 - Uniform
- Unusual Features
 - Gaps
 - Outliers

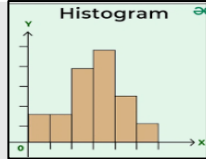
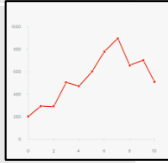
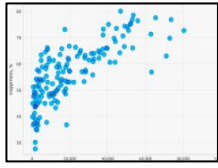
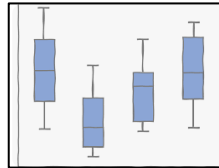
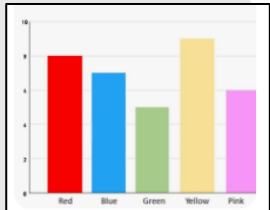
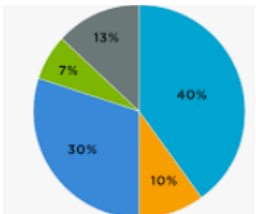
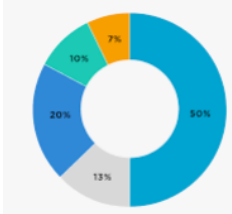
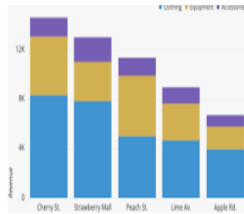
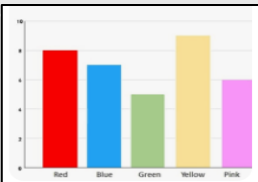
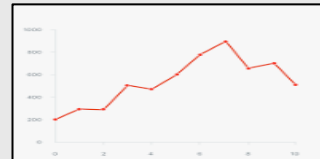
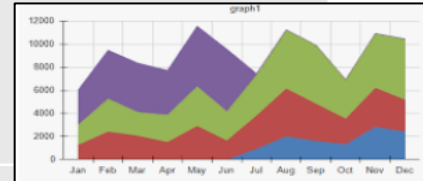
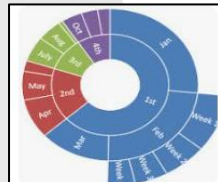
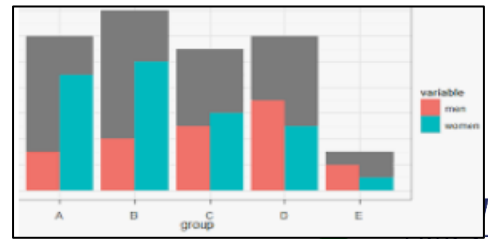


How to Choose a Right Graph for Data Visualization



Common data types and corresponding chart types

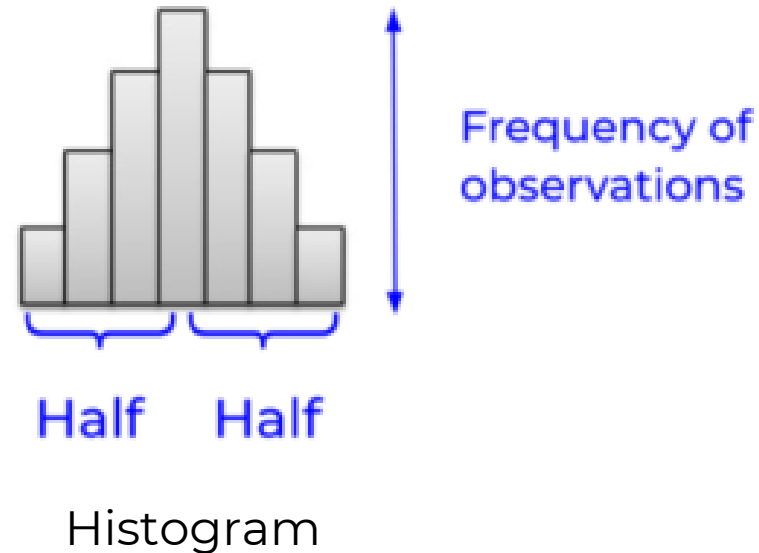


Data Type	Recommended Chart Types
Numeric	Histogram, Line Chart, Scatter Plot, Box Plot    
Categorical	Bar Chart, Pie Chart, Donut Chart, Stacked Bar Chart    
Ordinal	Bar Chart, Dot Plot, Heatmap   
Time Series	Line Chart, Area Chart, Heatmap   
Hierarchical	Treemap, Sunburst Chart, Nesting Chart   

Graphical Representation of Data

Center

- **The center** of the distribution is graphical at the **median** of the distribution.
- **Half** of the observations are on both sides.
- The **height** of the column indicates the **frequency of observations**.



Center

Spread

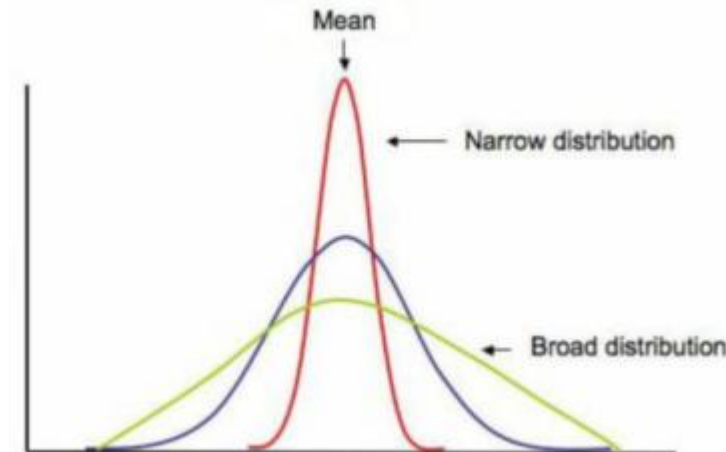
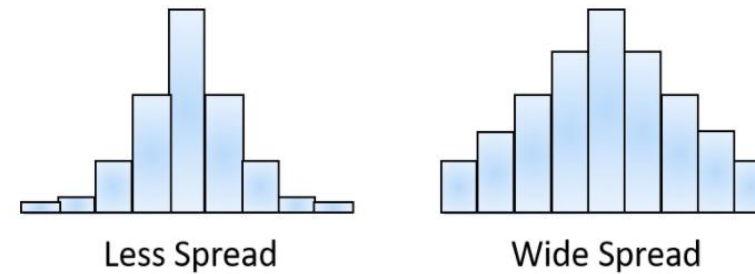
Shape

Unusual
Fea.

Graphical Representation of Data

➤ Spread

- Variation of data
- If the set of observations spans a wide range
- If observations are centered around a single value over a narrower range.....



Center

Spread

Shape

**Unusual
Fea.**

Normally Distribution Videos

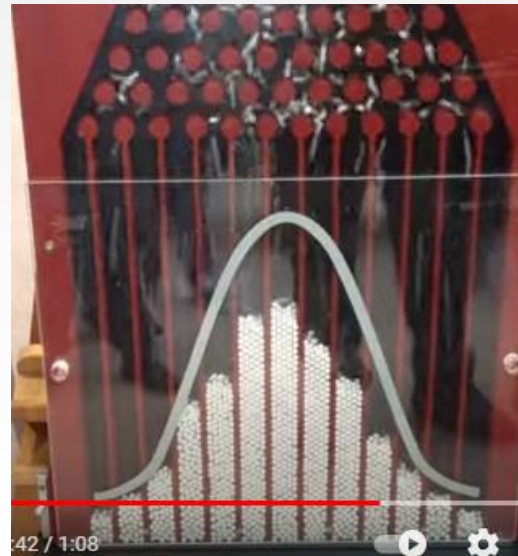
▶ Video-1

- <https://www.youtube.com/watch?v=Bampgm0HKDU>

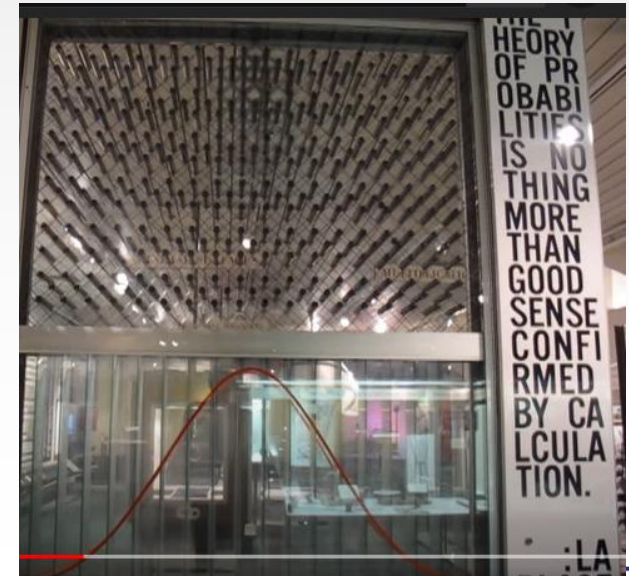


▶ Video-2

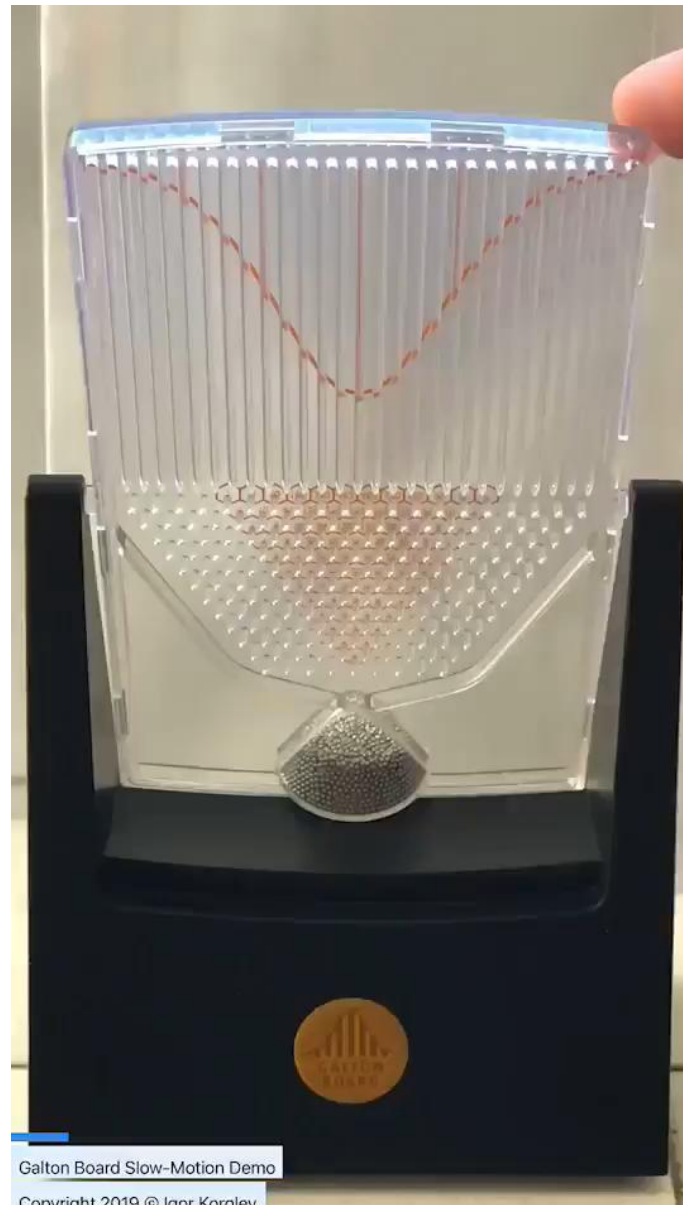
- <https://www.youtube.com/watch?v=4HpvBZnHOVI>



▶ Video-3

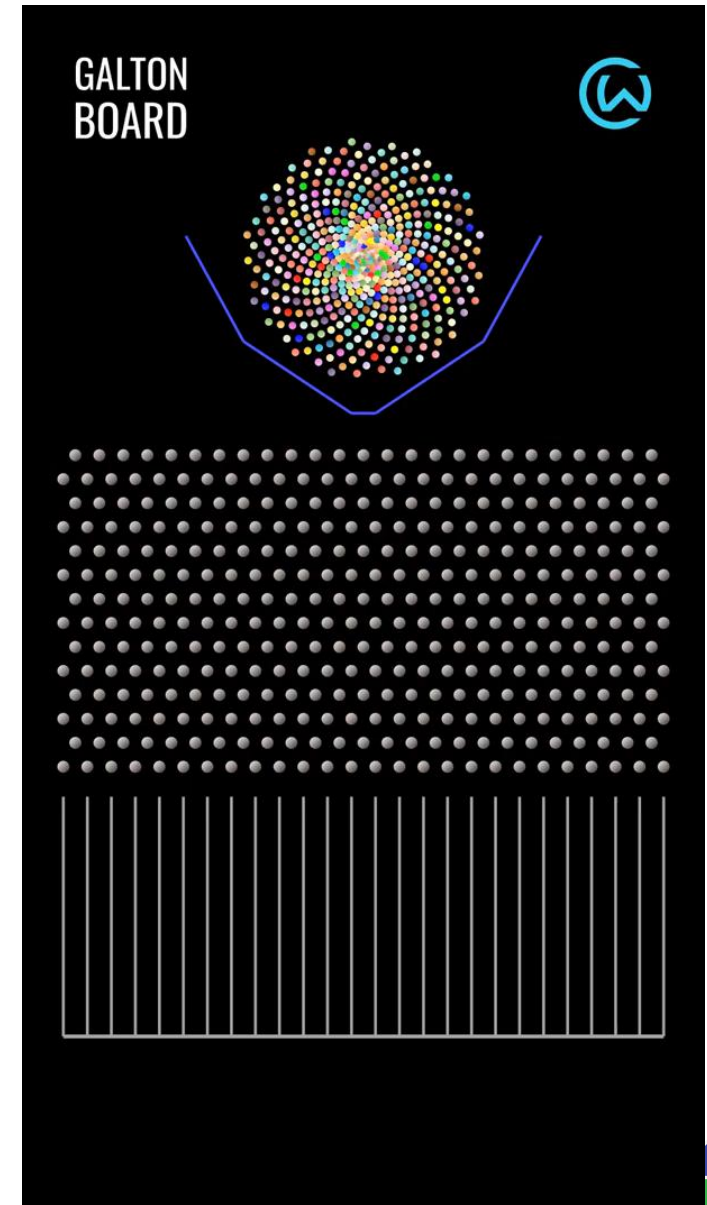


Normal distribution is everywhere..



Galton Board Slow-Motion Demo

Copyright 2019 © Igor Korolev

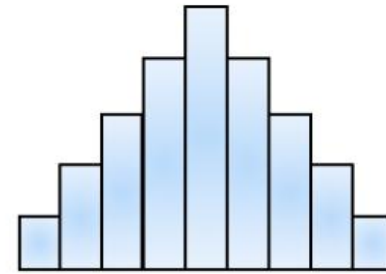


Graphical Representation of Data

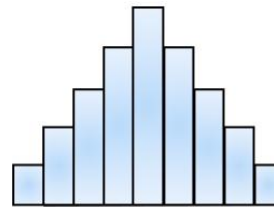
Shape

The shape of a distribution can be defined using the following properties.

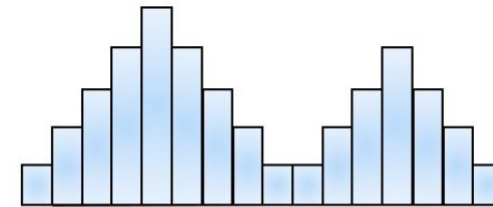
- Symmetric
- Number of Peaks
- Skewness
- Uniform



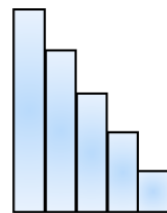
Symmetric



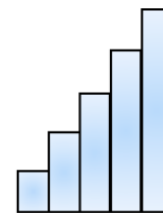
Unimodal,
Bell Shaped



Bimodal



Skewed Right



Skewed Left



Uniform

Center

Spread

Shape

**Unusual
Fea.**

Probability distributions

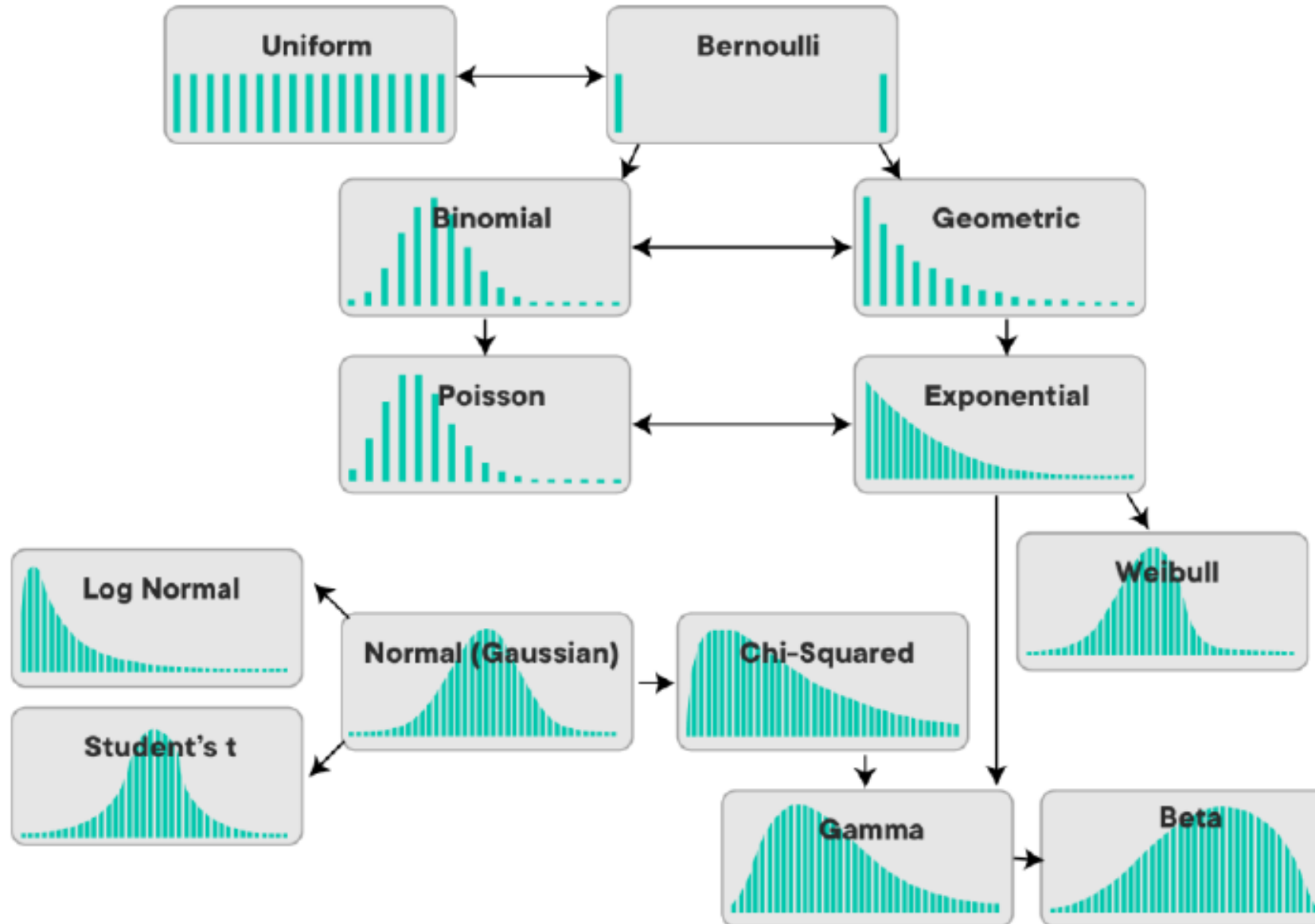
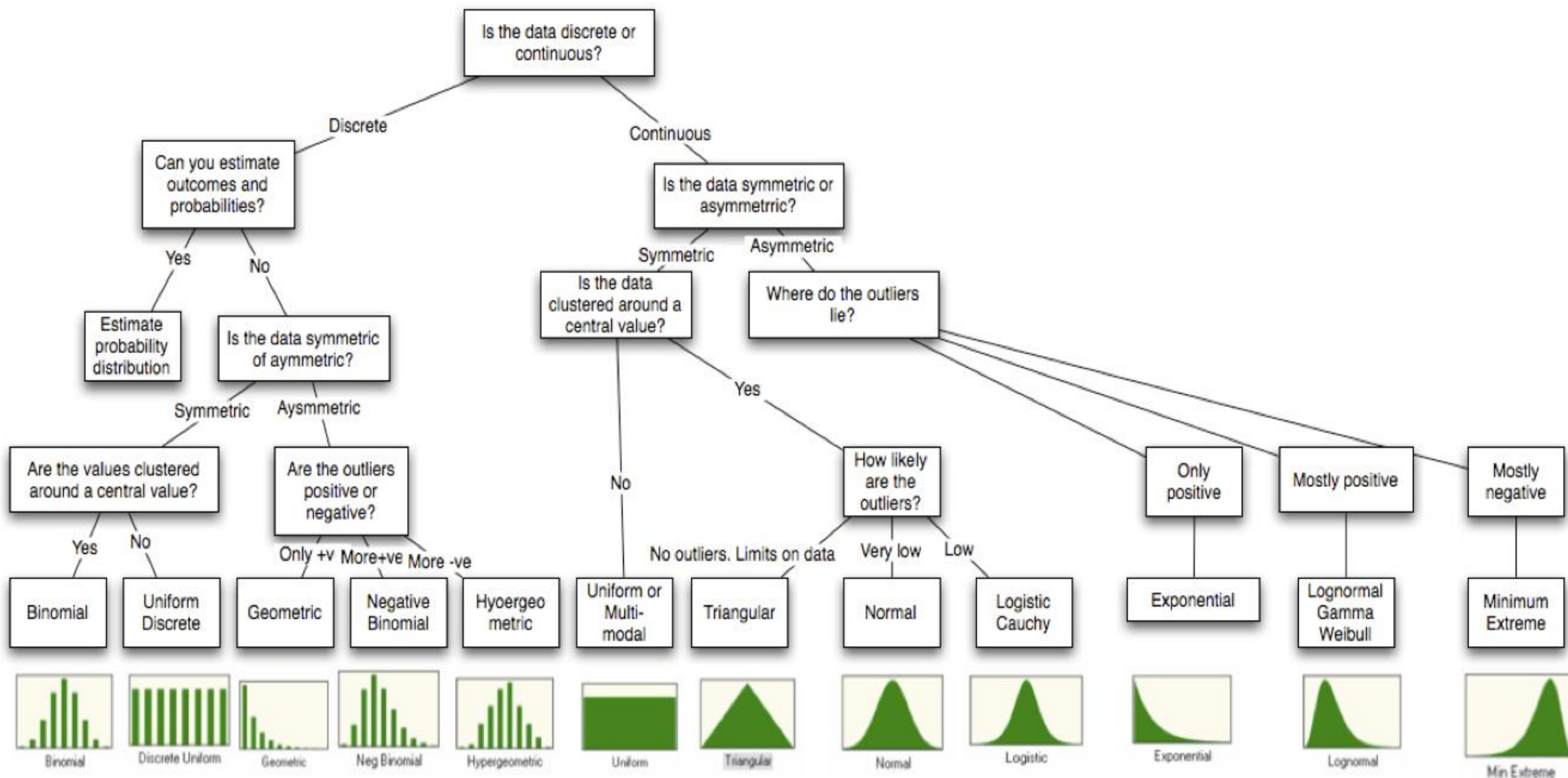


Figure 6A.15: Distributional Choices

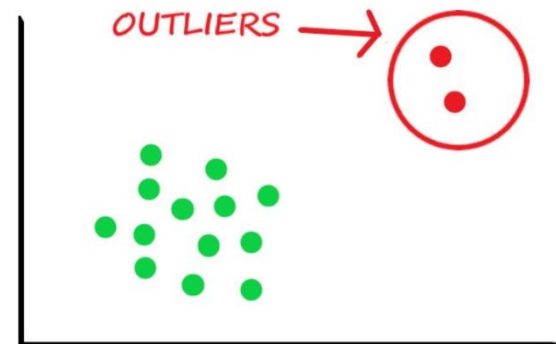
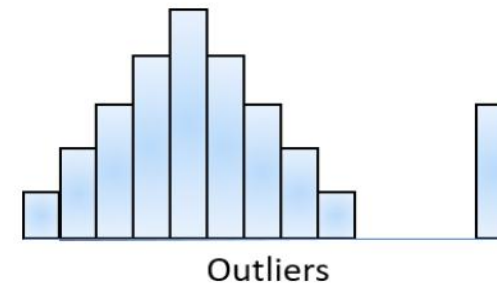
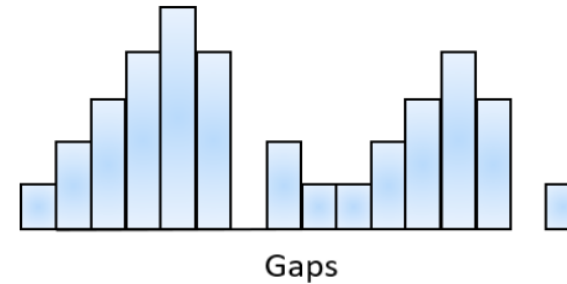


Graphical Representation of Data

► Unusual Features

Common unusual features of data models are:

- Gaps
- Outliers



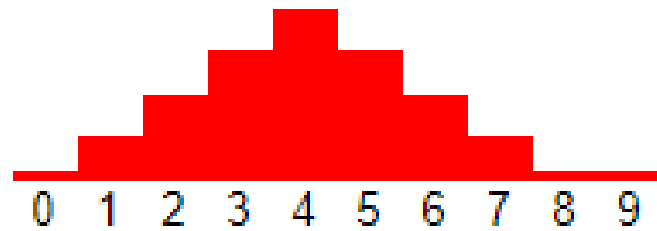
Center

Spread

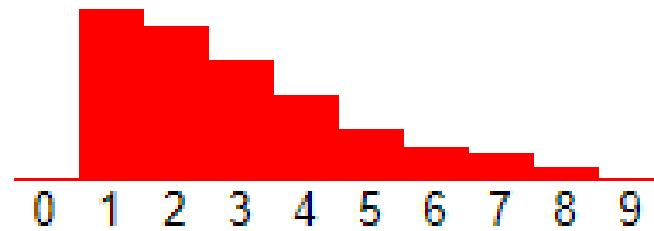
Shape

**Unusual
Fea.**

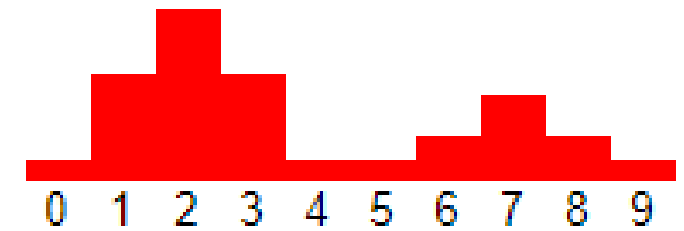
Data Patterns



Symmetric, unimodal,
bell-shaped



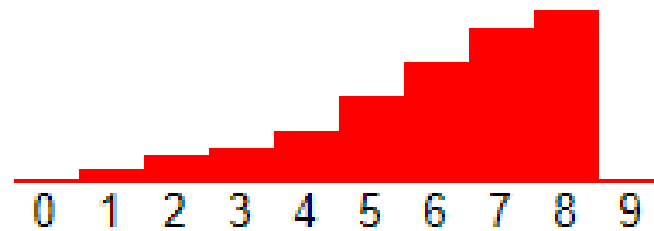
Skewed right



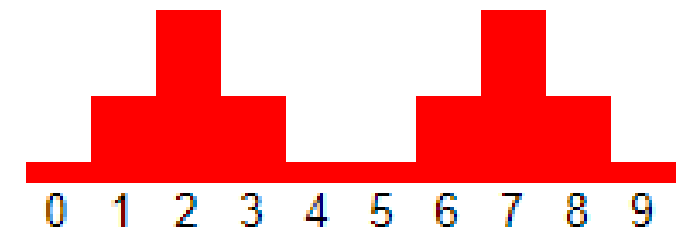
Non-symmetric, bimodal



Uniform



Skewed left

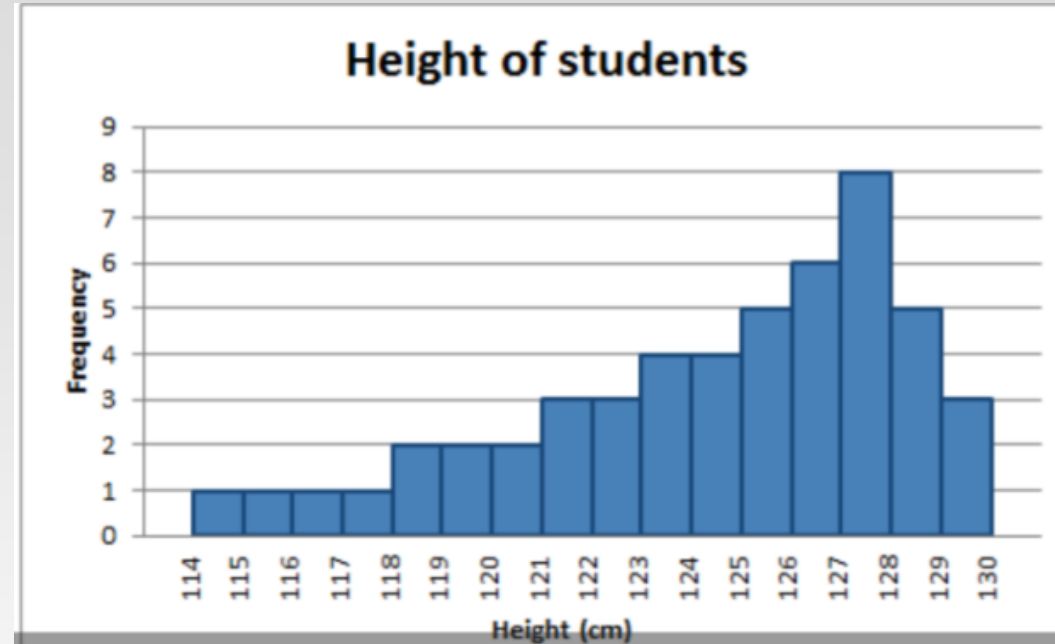


Symmetric, bimodal



Pear Deck

TEST



Which pattern is correct for this figure?

- a. Right-skewed with no outliers
- b. Right-skewed with one outliers
- c. Left-skewed with no outliers
- d. Symmetric

Task -2

Task 2: Create a frequency histogram from the data in the table below. What you can conclude about the shape of the distribution?

Income (In thousands of dollars)	Number of families
16-22	2
23-29	3
30-36	5
37-43	8
44-50	8
51-57	10

Excel

Frequency

Methods used in descriptive statistics:

- Frequency Tables
- Shapes and Graphics
- Histogram and Frequency Polygons
- Column and Pie Charts

Developer Type	Frequency	Relative Frequency
Front-end Developer	25	0.25
Backend Developer	15	0.15
Full-stack Developer	20	0.20
Data Scientist	40	0.40

Intervals	Classes	Frequency	Frequencies
	1-4	4	
	5-8	5	
	9-12	3	
	13-16	4	
	17-20	2	

Frequency

➤ Frequency

- Number of occurrences of a data value

➤ Relative Frequency

- How often something happens divided by all outcomes

➤ Cumulative Frequency

- Accumulation of previous relative frequencies

DATA VALUE	FREQUENCY
3	5
4	3
5	6
6	2
7	1

DATA VALUE	FREQUENCY	RELATIVE FREQUENCY
2	3	$\frac{3}{20}$ or 0.15
3	5	$\frac{5}{20}$ or 0.25
4	3	$\frac{3}{20}$ or 0.15
5	6	$\frac{6}{20}$ or 0.30
6	2	$\frac{2}{20}$ or 0.10
7	1	$\frac{1}{20}$ or 0.05

DATA VALUE	FREQUENCY	RELATIVE FREQUENCY	CUMULATIVE RELATIVE FREQUENCY
2	3	$\frac{3}{20}$ or 0.15	0.15
3	5	$\frac{5}{20}$ or 0.25	$0.15 + 0.25 = 0.40$
4	3	$\frac{3}{20}$ or 0.15	$0.40 + 0.15 = 0.55$
5	6	$\frac{6}{20}$ or 0.30	$0.55 + 0.30 = 0.85$
6	2	$\frac{2}{20}$ or 0.10	$0.85 + 0.10 = 0.95$
7	1	$\frac{1}{20}$ or 0.05	$0.95 + 0.05 = 1.00$

QUESTION

What is the proportion of dogs living up to 12 years (at most)?

Age	Frequency	Relative Frequency	Cumulative Relative Frequency
0	2	$\frac{2}{19}$	0.1053
2	3	$\frac{3}{19}$	0.2632
4	1	$\frac{1}{19}$	0.3158
5	3	$\frac{3}{19}$	0.4737
7	2	$\frac{2}{19}$	0.5789
10	2	$\frac{2}{19}$	0.6842
12	2	$\frac{2}{19}$	0.7895
15	1	$\frac{1}{19}$	0.8421
20	1	$\frac{1}{19}$	1.0000

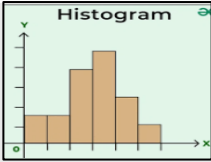
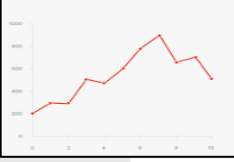
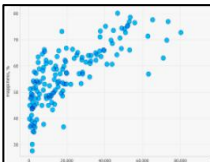
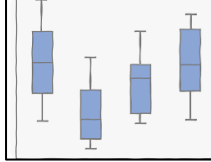
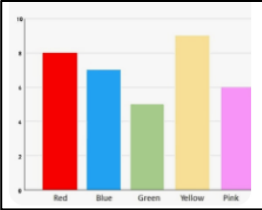
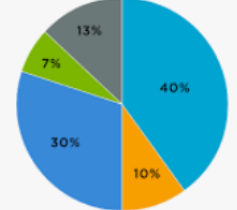
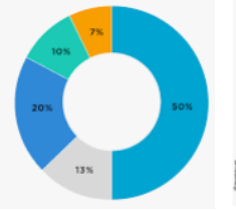
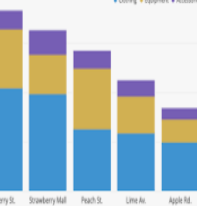
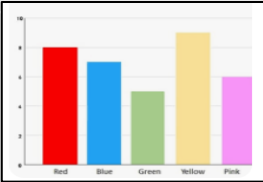
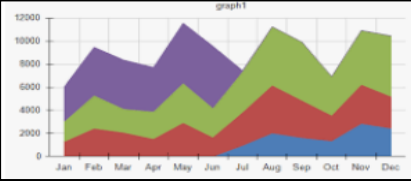
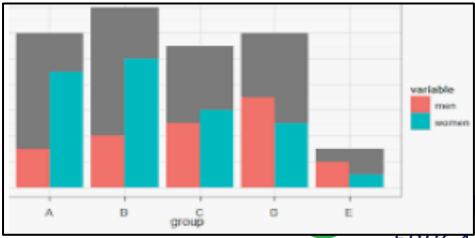
Graphs and Charts

► Why Charts ?

- Makes more understandable/readable.
- Points of interest are indicated.
- Provides information about the shape of the distribution.
- Prediction becomes easier.

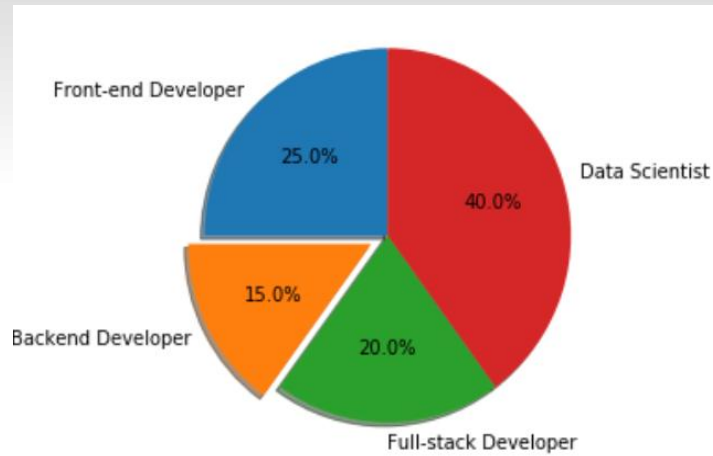


Common data types and corresponding chart types

Data Type	Recommended Chart Types
Numeric	Histogram, Line Chart, Scatter Plot, Box Plot    
Categorical	Bar Chart, Pie Chart, Donut Chart, Stacked Bar Chart    
Ordinal	Bar Chart, Dot Plot, Heatmap   
Time Series	Line Chart, Area Chart, Heatmap   
Hierarchical	Treemap, Sunburst Chart, Nesting Chart   

Pie Charts

- For **categorical data**!
- Usually used for **nominal** and **ordinal** variables.
- The circle is sliced into pie slices to complete 100% of the total.
- Each slice presents the attribute of the variable.

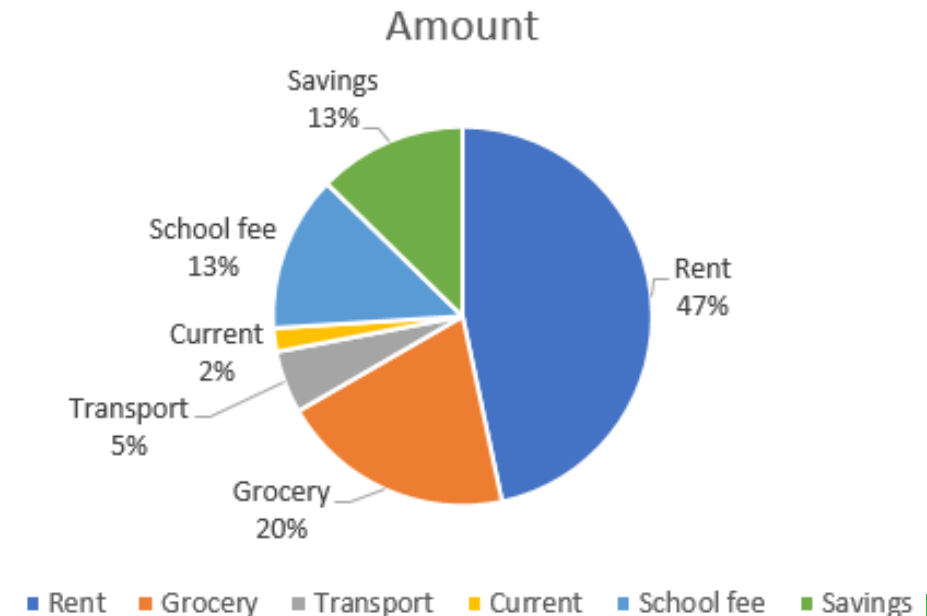


Pie Chart



Pie Chart Examples

1	Expenses	Amount
2	Rent	7000
3	Grocery	3000
4	Transport	800
5	Current	300
6	School fee	2000
7	Savings	1900
8		
9		
10		
11		
12		
13		

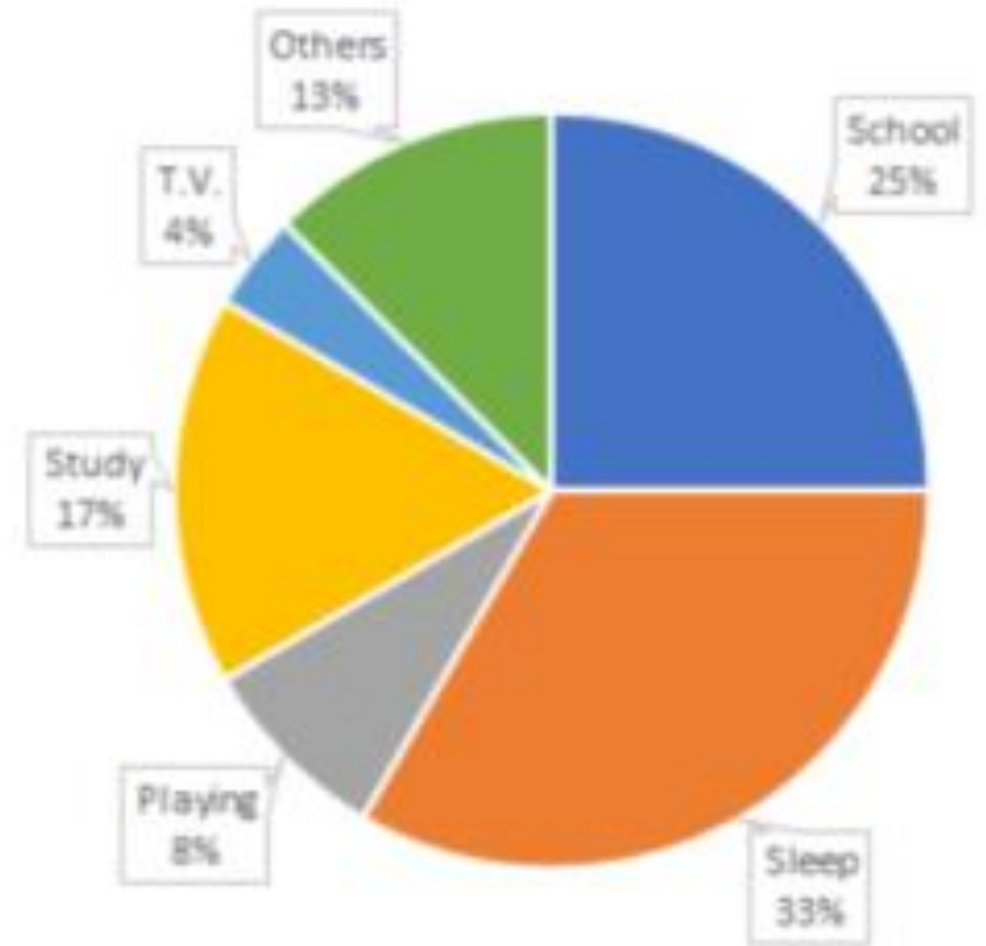


Pie Chart

Example

- Calculating Pie Percentage

Activity	No. of Hours	Measure of central angle
School	6	$(\frac{6}{24} \times 360)^\circ = 90^\circ$
Sleep	8	$(\frac{8}{24} \times 360)^\circ = 120^\circ$
Playing	2	$(\frac{2}{24} \times 360)^\circ = 30^\circ$
Study	4	$(\frac{4}{24} \times 360)^\circ = 60^\circ$
T. V.	1	$(\frac{1}{24} \times 360)^\circ = 15^\circ$
Others	3	$(\frac{3}{24} \times 360)^\circ = 45^\circ$



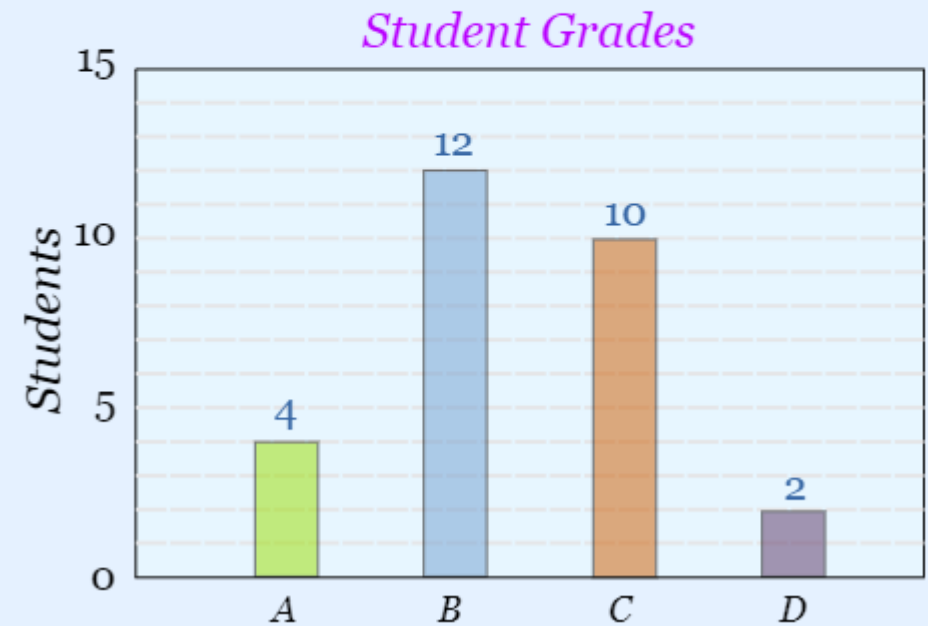
Bar Charts

- For **categorical** data!
- Usually used with **nominal** and **ordinal** variables.
- Each of the bars (columns) represents different values of a variable.
- Each bar's height indicates the frequency of each attribute.



Grade:	A	B	C	D
Students:	4	12	10	2

bar graph:

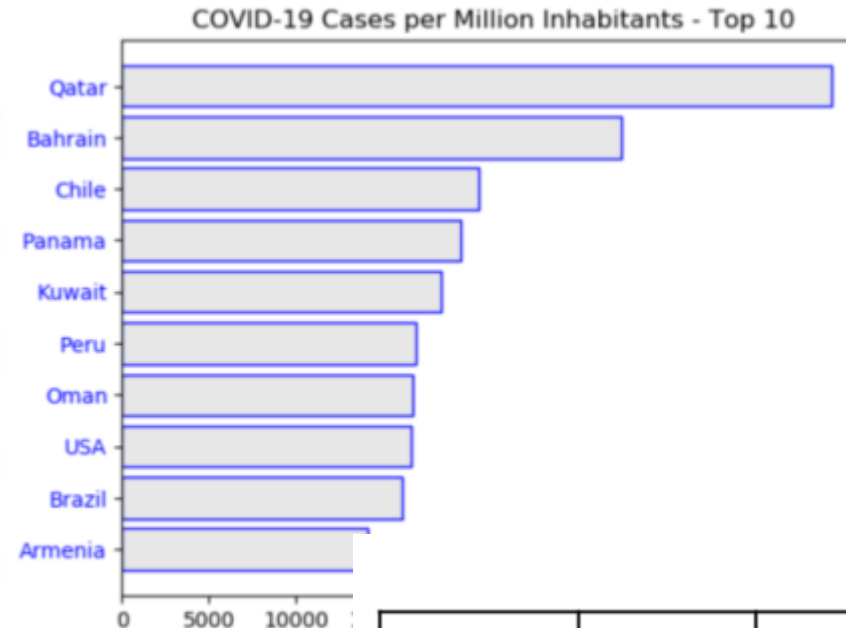


Bar Charts

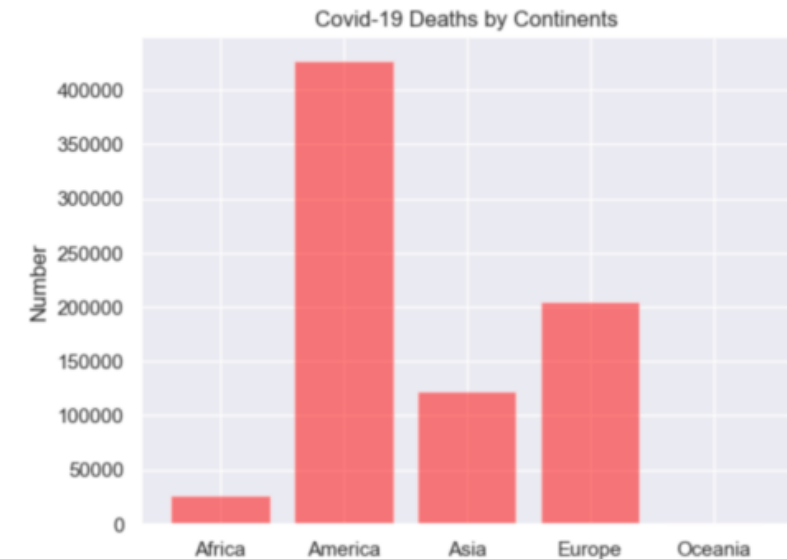
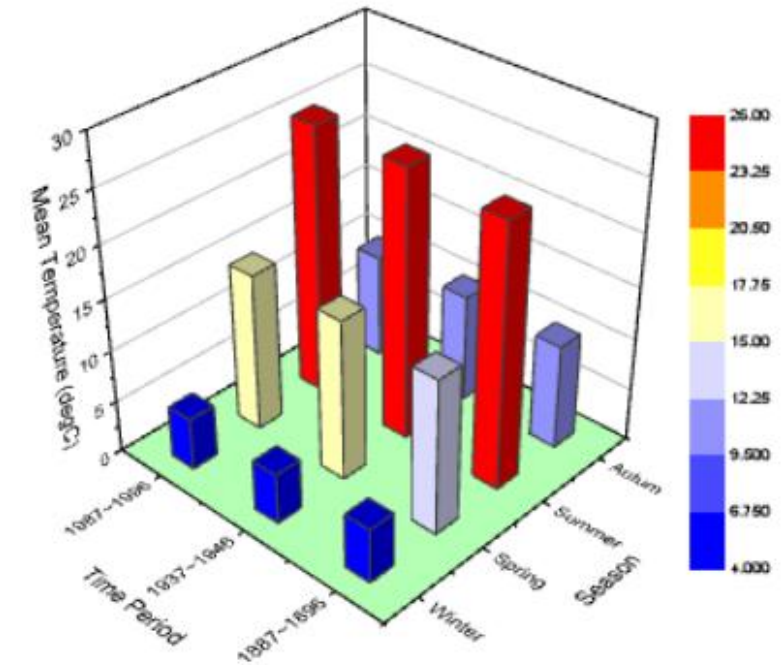
3D

Seasonal Mean Temperature with Standard Deviation

	cases	deaths	popData2019	casesPer1M
countriesAndTerritories				
Qatar	115661	193	2832071.0	40839.724710
Bahrain	47185	175	1641164.0	28750.935312
Chile	388855	10546	18952035.0	20517.849402
Panama	82790	1809	4246440.0	19496.331044
Kuwait	77470	505	4207077.0	18414.210151
Peru	549321	26658	32510462.0	16896.745423
Oman	83418	597	4974992.0	16767.464149
USA	5482416	171821	329064917.0	16660.591016
Brazil	3407354	109888	211049519.0	16144.808177
Armenia	41846	832	2957728.0	14148.021725



continent	cases	deaths
Africa	1119579	26260
America	11698368	427207
Asia	5606210	122034
Europe	3239237	205144
Oceania	25742	471



Task -3

Task 3: The number of passengers of an airline company by years is given in the table below. Create a bar chart based on these data.

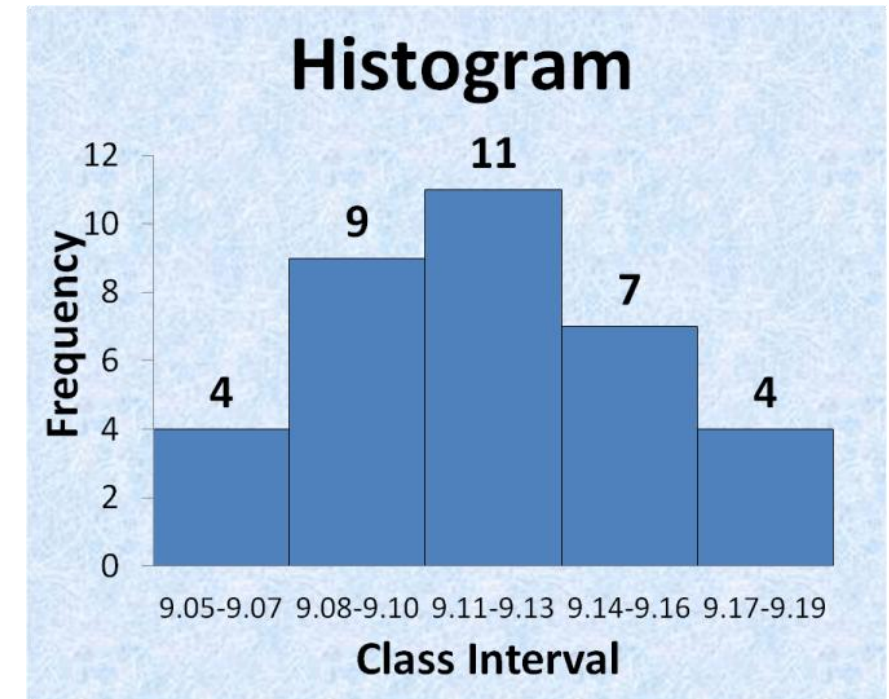
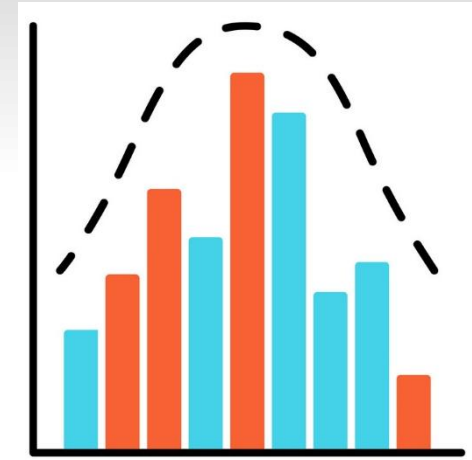
The number of passengers per year

Years	2010	2012	2013	2014	2015	2016	2017	2018	2019
Number of passengers (x1000)	5	7	13	10	20	22	17	16.5	27

Excel

Histogram

- Used with **Interval** or **Ratio** variables.
- Represents the frequency of each attribute for a variable.
- Gives you a good bird's eye view of the distribution of your data.



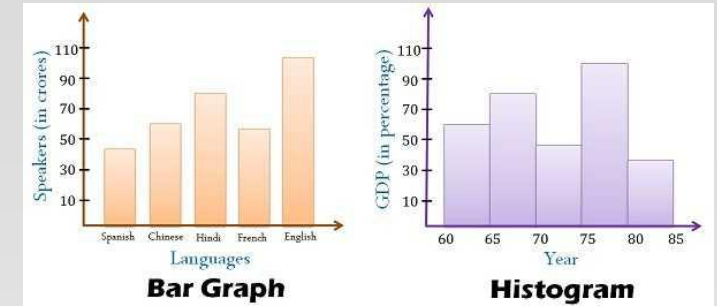
Bar Chart vs. Histogram

➤ Bar Chart

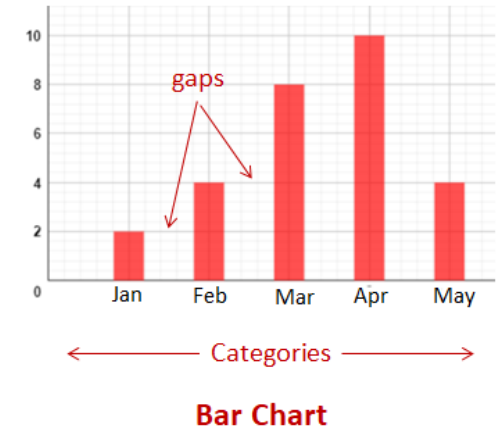
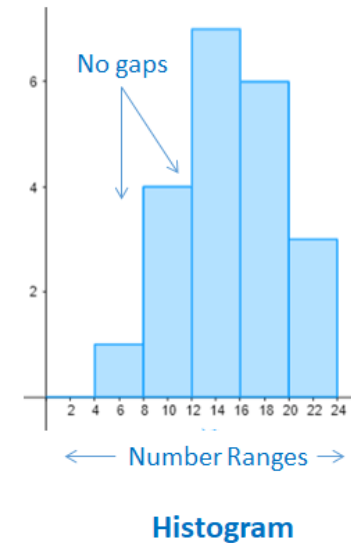
- Categories are present.
- A schematic comparison of discrete variables.
- Provides categorical data.
- Gaps between bars.

➤ Histogram

- Refers to graphic representation.
- Frequency distribution of continuous variables
- Provides numerical data.
- No Gaps between bars.



Histogram vs. Bar Chart



Line Chart

- Useful for **continuous numerical** data!
- Especially for time series.
- We put time to x-axis.
- Values locate at y-axis.

